

Genotyping (*Arabidopsis*)

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gDNA Extraction

- 1) Add 30 μL of TPS solution to an autoclaved 1.5-mL microcentrifuge tube
- 2) Add approximately 100 mg of leaf tissue (about the size of a small fingernail)
- 3) Grind the tissue thoroughly using a 1,000- μL pipette tip (or an autoclaved pestle)
- 4) Incubate at 70°C for 25 min
- 5) Add 40 μL of autoclaved distilled water (ADW)
- 6) Vortex thoroughly
- 7) Store at -20°C until PCR analysis
- 8) Vortex the sample, centrifuge at 13,000 rpm for 1 min, and use the supernatant as the DNA template

PCR Mixture

Component	Volume per reaction
2× ThumbTaq PCR Pre-Mix	10 μL
Genomic DNA	1 μL
Primer mix (10 pmol/ μL ; DW:Forward:Reverse = 90:5:5)	2 μL
Autoclaved distilled water (ADW)	7 μL
Total	20 μL

Reagent: 2× ThumbTaq PCR Pre-Mix

https://bio-ft.com/product/product.html?category=015024026028&id=TT116-500&mode=view&sort_mode=

PCR Cycle

Initial denaturation: 95°C for 2 min

Amplification (40 cycles): 95°C for 20 sec / 60°C for 40 sec / 72°C for 1 min/kb

Final extension: 72°C for 5 min

*TPS solution

Component	Stock concentration	Final concentration	Volume
Tris-HCl, pH 7.5	1 M	200 mM	10.0 mL
NaCl	5 M	250 mM	2.5 mL
EDTA, pH 8.0	0.5 M	25 mM	2.5 mL
SDS	10% (w/v)	0.5%	2.5 mL
ADW			Bring to 50 mL